

TinyGo has 2 features: PathFinder and Chunk

So how they work?

PathFinder uses Godot NavigationAgent3D to find the nearest path to the target. It returns the Vector3 of the target Location and then you can apply the position of your enemy or anything you want it to go to a target point with the given path from PathFinder.



Chunk uses tscn scene file as Chunks to load on given distance and unload when out of range. Chunk also has preload feature that it instantiates on given distance of preload distance before reaches the distance to make the chunk visible. So that makes sure that the chunk is ready before the player reached the visible distance.



Example code:

PathFinder:

```
// Properties
[Export] public NavigationAgent3D agent;
[Export] public Node3D target;
[Export] public float speed = 10.0f;
public Pathfinder _Agent = new Pathfinder();

1 reference
public override void _Ready()
{
    base._Ready();
    // Settings
    _Agent.Self = this;
    _Agent.Target = target;
    _Agent.Agent = agent;

    // Optional
    // _Agent.DesireDistance = 1.0f;
    // _Agent.TargetDesireDistance = 1.0f;

    // Init
    _Agent.Init();
}

1 reference
public override void _PhysicsProcess(double delta)
{
    base._PhysicsProcess(delta);
    LookAt(target.Position);
    // Update Path
    _Agent.Update();
    // Go to the Updated Path
    Velocity = _Agent.Direction_To_Target * speed;

    MoveAndSlide();
}
```

Code above shows the example to setup and apply the position to go to the target position.

Target can be anything you want as the location you want it to go. This can be useful for enemy to chase player that it doesn't walk blindly and hit the wall.

Remember to always add NavigationAgent3D to the chaser node and always bake the NavigationRegion3D to make it works.

Chunk:

```
7 references
internal partial class chunks : Node
{
    [Export] public PackedScene chunk;
    [Export] public CharacterBody3D player;
    [Export] public Node3D location;
    public Chunk _Chunk = new Chunk();

    1 reference
    public override void _Ready()
    {
        _Chunk.chunk = chunk;
        _Chunk.Player = player;
        _Chunk.LocationNode = location;
        _Chunk.ChunkDistance = 100.0f;

        // Enable Preload and the Preload distance must bigger than the Distance
        // Enabling Preload is optional but is recommended to enable
        _Chunk.EnablePreLoad_Chunk = true;
        _Chunk.PreLoad_Distance = 150.0f;
    }

    1 reference
    public override void _Process(double delta)
    {
        _Chunk.Update();
    }
}
```

This code above shows the example of making a chunk instance. Using Chunk in TinyGo is one tscn chunk per instance so when you have multiple chunks, you need to add new Chunk instance for them. This ensures the chunk is independent and not crashing other. The LocationNode is where the chunk will appear when the player is within the given distance.

Enabling Preloading is recommended because it makes sure the chunk is ready before the player enters the distance range.